

Muhammad Danish

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Education

University of New Mexico

- Ph.D. in Computer Science
- M.S. in Computer Science
- B.S. in Computer Science; Minor in Honors Interdisciplinary Liberal Arts

Albuquerque, NM

Aug 2025 – May 2028

Aug 2025 – Jul 2026

Aug 2021 – May 2025

Research & Publications

1. **Danish, M., et al. “[The Tragedy of Convenience: Cascading User-Data Leakage from SMS-Delivered URLs](#).”**
ACM Conference on Computer and Communications Security (CCS), 2026.
Led a large-scale study of 33M+ SMS messages and 322K URLs; built an end-to-end LLM-assisted pipeline to identify 170+ PII-exposing services; Performed token-entropy analysis, bounded enumeration, authentication testing, API and WebSocket analysis, and coordinated disclosure with remediations protecting 120M+ users.
2. **Sobrados, E., Danish, M., et al. “[Exposing Probe-Resistant VPNs via Protocol Coexistence](#).”**
IEEE Symposium on Security and Privacy (SP), 2027.
Developed an Internet-measurement pipeline that reverse-engineered commercial VPN applications and enumerated 72,887 servers; found that 83.4% were fingerprintable because probe-resistant protocols coexisted with detectable legacy or misconfigured protocols.
3. **Danish, M., et al. “[View-Based Monetization Fraud Across Content-Sharing Platforms](#).”** (Submitted)
Led a cross-platform study of view inflation across a diverse set of monetizable content-sharing services (e.g., Spotify, Dailymotion, Ameblo); 9 of 10 platforms counted more than 80% of attempted views and attacks on five platforms persisted & scaled using consumer-grade infrastructure;
4. **Danish, M., et al. “[A Cross-Attack Security and Privacy Evaluation of Model Compression](#).”** (Submitted)
Systematically evaluating how quantization, pruning, knowledge distillation, low-rank factorization, and hybrid compression affect model security and privacy across evasion, poisoning, and privacy attacks under varying compression levels, datasets, and architectures.
5. **Danish, M., et al. “[Robustness and Security of Embedded Edge AI for Space Systems](#).”** (In Progress)
Evaluating deployment and security of AI models across FPGA-based and GPU-based edge platforms across space-specific threats such as radiation-induced faults, adversarial fault injection, resource exhaustion, and model corruption under constrained operating conditions.

Experience

- **Research Assistant (PI and Reference: [Afsah Anwar](#)), University of New Mexico** Dec 2023 – Present
 - Conducted large-scale AI, security & privacy research, resulting in multiple publications.
 - Built end-to-end pipelines and reverse-engineered AI/web/mobile systems to uncover vulnerabilities.
 - Characterized security & privacy issues in multiple systems (e.g., web, AI/ML) and proposed mitigation strategies.
- **Student Programming Analyst, University of New Mexico** Oct 2022 – Aug 2025
 - Automated cross-department administrative workflows using Microsoft Power Platform, Power Automate, and Boomi.
 - Built regression-test suites and integrated them into Azure DevOps CI/CD pipelines, reducing testing time by ~50%.
 - Delivered web applications and system integrations across multiple departments, supporting 20,000+ students.
- **Computer Science Teaching Assistant, University of New Mexico**
 - Led weekly programming, networking, and security labs for 40+ students.
 - Developed exercises, graded assignments and examinations, and provided individualized technical feedback.
 - Fostered a collaborative learning environment via group discussions, one-on-one sessions, and [YouTube](#).
 - **Courses:** Introduction to Cybersecurity, Computer Networks, and Intro/Intermediate Programming

Skills

- **Languages:** Python (PyTorch and TensorFlow), Bash, PowerShell, Java, JavaScript, SQL, MATLAB
- **Technical:** adversarial ML, LLMs, agentic AI, model compression, web & API security, reverse engineering
- **Tools:** Linux/Windows, Playwright, Docker, NMap, Wireshark, Burp Suite, Kibana, Metasploit, Azure, Git

Awards, Media, and Volunteering

- **Awards:** [CRA Outstanding Undergraduate Researcher](#) (North America), NMEF Scholar (New Mexico), [Bug Bounty](#)
- **Media:** Research featured by [Ars Technica](#), [TechRadar Pro](#), [Help Net Security](#), [Azerbaijan CERT](#), and the [UK NCSC CTO](#); invited guest on [N2K CyberWire Research Saturday](#); profiled by [CRA Computing Research News](#).
- **Peer Review:** AEC member at top conferences e.g., IEEE S&P ([Award](#)), ACM CCS, NDSS, PETS ([Award](#))
- **Stanford Code in Place (Section Leader):** Mentored diverse students in Python and CS fundamentals.